Late Hadrynian and Helikian pole positions from the Spokane Formation, Montana

ICARO VITORELLO AND ROB VAN DER VOO
Department of Geology and Mineralogy, The University of Michigan, Ann Arbor, MI, U.S.A. 48109

Received 25 May 1976
Revision accepted for publication 10 September 1976

The natural remanent magnetizations of Beltian argillites from the Spokane Formation (formerly Grinnell Formation) have been analyzed with alternating field and thermal demagnetization techniques. Two groups of magnetic directions have been observed. One is assumed to be the primary magnetization, representative of the Helikian (approx. 1300 Ma) geomagnetic field: D = 201.5°; I = +40°; k = 61; and α_{95} = 5.1° for N = 14 sites (51 samples), yielding a pole position at 15.5 °S, 134.5 °W, dp = 3.7°, dm = 6.2°. Comparisons with contemporaneous pole positions reveal good agreement. The other direction of magnetization corresponds to a Late Hadrynian age (approx. 600 to 750 Ma) and resulted from low-grade metamorphism. It is speculated that this thermal event resulted from enhanced heat flow during continental breakup in the Late Hadrynian. This direction (D = 267.5°; I = +17°; k = 10; α_{95} = 9.1° for N = 28 specimens) yields a pole position at 5 °N, 163.5°E, with dp = 4.9°, dm = 9.4°. Both groups of directions of magnetization are pre-folding, which occurred during the Laramide orogeny. Laramide deformation was only moderate, however, and the eastward thrusting of the Beltian sediments over the Lewis thrust was probably not accompanied by significant structural rotation.

Les aimantations remanentes naturelles des argillites beltiennes de la "Spokane Formation" (auparavant Grinnell Formation) ont été analysées utilisant des techniques de désaimantation thermique et des champs alternatifs. Deux groupes de directions d'aimantation ont été observés. L'un est considéré comme l'aimantation primaire, représentant le champ géomagnétique hélikien (daté d'environ 1300 Ma) : D = 201.5°; I = +40°; k = 61, et α_{95} = 5.1° pour N = 14 sites de prélèvement (51 échantillons). Cette direction a fourni un pôle géomagnétique à 15.5°S, 225.5°E, dp = 3.7°, dm = 6.2°. Ce pôle s'accorde bien avec d'autres pôles du même âge. L'autre direction d'aimantation correspond à un âge hadrynien supérieur (environ 600 à 750 Ma) et cette direction est le résultat d'un métamorphisme léger. Nous proposons que ce métamorphisme thermique est le résultat d'un flux géothermique élevé ayant lieu pendant une rupture continentale hadrynienne. Cette seconde direction (D = 267.5°; I = +17°; k = 10; et α_{95} = 9.1° pour N = 28 spécimens) a fourni un pôle géomagnétique à 5°N, 163.5°E, avec dp = 4.9° et dm = 9.4°. Les deux groupes d'aimantations ont précédé le plissement Laramide. Cependant, la déformation Laramide était légère et le charriage vers l'est des sédiments beltiens au dessus de la faille Lewis n'était probablement pas accompagné d'une rotation structurelle considérable.

Can. J. Earth Sci., 14, 67–73 (1977)

Introduction

A thick sequence of limestones and argillites of the Belt Supergroup is well exposed in Glacier National Park, Montana. Several formations with argillites display a striking red color and we report here the results of 51 paleomagnetic samples that were collected at 14 sites in one of these units, the Spokane Formation (Fig. 1). We follow here the name recommended by Harrison (1972), but note that the name Grinnell Formation was formerly in use.

Paleomagnetic studies of the Spokane Formation of Montana and British Columbia were previously published by Collinson and Runcorn (1960), Black (1963), and Runcorn (1964). The pole positions determined from these studies were not obtained through demagnetization analyses, and since the redbeds are slightly metamorphosed we felt it necessary to reinvestigate.

The age of the formations of the Belt Supergroup is not well known. We follow Harrison (1972) in estimating the age of the Spokane Formation as roughly 1300 Ma old. This age estimate is based on radiometric ages (Obradovich and Peterman 1968), as well as several circumstantial arguments which will not be reiterated here. It must be noted, however, that an age some 200 Ma greater is not impossible (see also Evans *et al.* 1975, p. 58). Later deformation of the Beltian sediments has been moderate, the most important event being the eastward overthrusting of the Beltian over Cretaceous and older sediments during the Laramide orogeny (Ross 1959). Theoretically, the possibility exists that the overthrust mass has been structurally

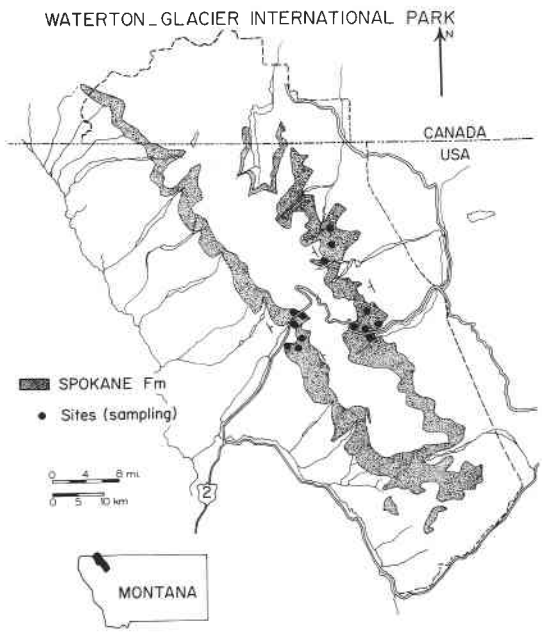


FIG. 1. Schematic geologic map of the Glacier National Park area, Montana. The fourteen sampling sites in the Spokane Formation are indicated.

rotated, though the geometry of the frontal Lewis thrust, the great thickness of the thrusted Beltian sediments, and the continuity of the Beltian outcrops over large distances along the strike argue against large rotations. Norris and Black (1961) concluded against relative rotations of thrust-sheet segments, on the basis of paleomagnetic studies.

Presently, the Spokane Formation in Glacier National Park is found well exposed because of post-Laramide uplift and erosion, particularly due to Quaternary glaciation. It forms a broad, cylindrical, synclinal structure. Of the 14 sites (Fig. 1), six had a northeasterly dip of no more than 25°, and eight had a southwesterly dip of no more than 20°. At least 800 m of section are represented in our samples.

Handsampling was required because of National Park regulations and, on the average, four handsamples were collected per site. Up to three cylindrical specimens were drilled in the laboratory from these samples.

After describing the results of the demagnetization analyses in the next section, we will discuss the directions of magnetization and the pole positions calculated from them. These data will be compared with other paleomagnetic results and evaluated in the light of pertinent geological observations from the literature on the Beltian rocks.

Measurement and Demagnetization

Measurement of the natural remanent magnetization (NRM) was carried out with a Schonstedt spinner magnetometer and with a ScT cryogenic magnetometer. The intensities of the NRM generally ranged between 1×10^{-3} Am^{-1} (1×10^{-6} emu/cm^3) and 20×10^{-3} Am^{-1}, though a few more strongly magnetic samples had intensities up to 70×10^{-3} Am^{-1}. The NRM directions are shown in Fig. 2a. They cluster in the third quadrant of the lower hemisphere, with a few directions scattered in the other quadrants, suggesting the presence of secondary magnetizations.

Both alternating field (AF) and thermal demagnetizations were carried out, in some cases consecutively, on the same specimens. Figure 3 shows the demagnetization diagrams obtained, plotted according to the Zijderveld (1967) method. AF demagnetization was first performed up to 100 milliTesla (1000 Oe), but generally failed to eliminate a significant portion of the NRM as can be seen for instance in Fig. 3a. Consequently, thermal treatment was required. Commonly, above temperatures of 660 °C the trajectory of further demagnetization was linear towards the origin. This last trajectory is interpreted as the removal of a single component of magnetization with a very high blocking temperature, residing in the fine-grained hematite of the argillites. This component, in view of its similarity in all samples, is referred to as the characteristic magnetization. The directions of the characteristic magnetizations, after unfolding, are shown in Fig. 2b; they cluster very

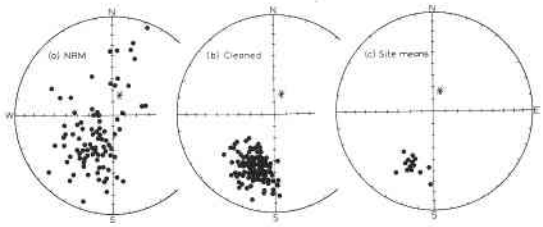


FIG. 2. Equal-area projections of the directions of magnetization of the Spokane Formation. Each dot represents the direction obtained from a specimen; in the projections of (b) and (c) the directions are plotted after thermal demagnetization above 660 °C and after tectonic correction. All symbols pointing downward; the asterisk is the present-day local geomagnetic field direction.

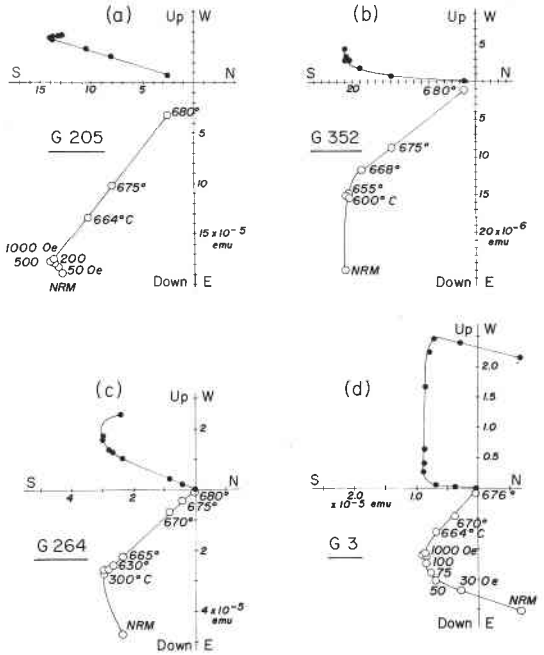


FIG. 3. Orthogonal projections of the successive end points of the magnetization vector during progressive AF and thermal demagnetization. Solid symbols represent projections on the horizontal plane, open symbols on the north-south vertical plane.

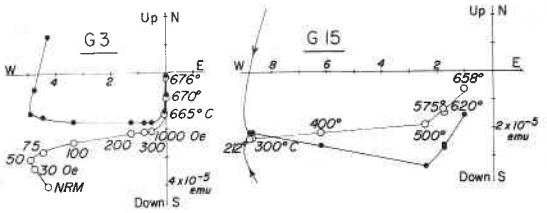


FIG. 4. Orthogonal projections of the successive end points of the magnetization vector during progressive demagnetization. For explanation of symbols see Fig. 3, though vertical projections here are on the east-west vertical plane in order to illustrate the nature of the intermediate magnetization in a clearer fashion.

well and are predominantly in the third quadrant of the lower hemisphere.

Generally, the directions of the soft components of magnetization conform to the local present-day geomagnetic field direction (Fig. 3a, 3c, or Fig. 3d below 50 Oe = 5 mT). However, yet another component of magnetization was found to be present in the specimen of Fig. 3d, of intermediate coercivity, and eliminated between 5 mT and 100 mT. Because visual recognition of this component is hampered by the way the demagnetization is plotted (up-down and east-west versus north-south) we have replotted the demagnetization diagram in Fig. 4. It can be seen that this magnetization has a westerly declination and a shallow, positive inclination; it is clearly different from the characteristic magnetization eliminated above 665 °C. Several other samples showed this same intermediate direction in thermal or AF demagnetization; another typical example is also shown in Fig. 4 (G 15), where the intermediate magnetization is eliminated in temperatures between 300 °C and 500 °C. It must be remarked that only vector demagnetization diagrams such as those of Fig. 4 ('Zijderveld plots') enable one to recognize multivectorial NRM as just described; they facilitate determination of the directions of individual components through vector subtraction. For this method straight trajectories, defined by three or more measurements, are a necessary condition to ensure that it is unlikely that two or more vector components are being removed at the same time. Most samples showing the intermediate direction of magnetization were demagnetized in sufficient detail to meet this condition.

The Characteristic Direction and its Pole Position

The mean directions of the characteristic magnetizations are summarized in Table 1. The results of individual specimens were averaged per sample, and the sample directions were averaged for each of the sites. These site mean directions are shown in Fig. 2c. They cluster very well after correction for the dip of the strata. The fold test of Graham (1949) is positive, since k increases from 21 to 61, whereas α_{95} decreases from 8.9° to 5.1° upon unfolding. Predictably, it does not make a difference whether the final mean direction is obtained by averaging sites, samples, or specimens (Table 1).

The mean direction yields a geomagnetic pole position at 15.5°S, 134.5°W, which is fairly close to the previous poles obtained from NRM data only (Collinson and Runcorn 1960; Black 1963; Runcorn 1964). More important is that this pole agrees well with contemporaneous pole positions, referred to as the 'Elsonian trend' during Helikian times (Spall 1971; see also Irving and McGlynn 1976). It is noteworthy that this group of poles represents results from several geological provinces of the Canadian Shield, as well as from Missouri and Colorado.

TABLE 1. Summary of the site mean directions of characteristic magnetization

Site	R_{sa}	R_{sp}	Before TC Decl.	Before TC Incl.	After TC Decl.	After TC Incl.	α_{95}	k
1	2/2	(4/5)	212.1	+34.3	202.6	+46.0	—	57
2	1/1	(2/2)	214.7	+14.9	212.6	+27.5	—	—
3	5/5	(8/8)	195.6	+20.9	190.8	+32.7	12.1	41
4	3/4	(6/8)	192.7	+46.7	200.4	+40.7	17.0	54
5	5/5	(9/9)	193.3	+63.0	205.2	+50.2	10.9	50
6	4/4	(7/7)	196.3	+41.5	204.7	+43.8	11.1	69
7	5/5	(7/8)	200.3	+51.3	206.0	+43.0	15.4	26
8	4/4	(6/7)	195.1	+47.2	202.5	+40.9	6.2	224
9	3/3	(6/6)	187.2	+8.8	181.9	+27.8	25.5	25
10	4/4	(8/8)	208.8	+24.6	204.8	+35.0	20.7	21
11	3/3	(6/6)	202.6	+47.8	209.5	+40.7	9.5	171
12	5/5	(10/10)	203.8	+59.3	216.1	+46.7	10.3	57
13	2/2	(4/4)	202.4	+51.0	207.9	+42.2	—	111
14	3/4	(5/7)	190.6	+36.8	183.8	+39.4	13.1	89
All specimens:		88/95			201.4	+40.3	2.8	31
All samples:	49/51				201.5	+41.1	3.6	33
All sites, before tectonic correction:		14/14	199.9	+39.5			8.9	21
All sites, after tectonic correction:		14/14			201.7	+40.2	5.1	61

NOTE: R_{sa} is the ratio of samples used for computation of the mean and its statistical parameters to the number of samples collected; R_{sp} is the corresponding ratio for specimen used to specimen measured; TC is the correction for the dip of the strata.

The Intermediate Direction and its Pole Position

Progressive AF and thermal demagnetizations showed the presence of a component of intermediate coercivity or blocking temperature ranges (Fig. 4). This phenomenon was only recognized at a late stage of data analysis; several samples were already demagnetized in a partially progressive fashion primarily to isolate the high-temperature magnetization, and possible information about the intermediate components could not be retrieved. A total of 28 specimens, however, showed the intermediate component and the directions obtained through vector subtraction are listed in Table 2 and shown in Fig. 5. Close inspection of previous publications, moreover, confirmed the presence of similar magnetizations in Beltian sediments (Black 1963: Kintla Formation, fig. 8; Runcorn 1964: Grinnell = Spokane Formation, fig. 6; Evans *et al.* 1975: Kintla Formation, table 1).

The mean direction, obtained by averaging the intermediate directions of these 28 specimens, is $D = 267.7°$, $I = +17.0°$, after unfolding. Again, the foldtest is positive at the 95 percent confidence level, k increasing from 6 to 10 and α_{95} decreasing from 12° to 9° ($N = 28$) upon unfolding; this same conclusion holds when site-means ($N = 11$) are used in the statistical analysis (Table 2). This mean direction yields a pole at 5°N, 163.5°E, dp = 4.9°, dm = 9.4°. It can be argued that the foldtest is barely conclusive and at most proves the intermediate magnetization to be pre-Laramide. However, the pole positions obtained from directions before and after unfolding (Table 2) are within a few degrees of each other, and both poles are very different from published Late Paleozoic and younger poles. A comparison between the pole position of the intermediate magnetization and the apparent polar wander path for cratonic North America suggests an Eocambrian (Late Hadrynian) age, since pole positions from the Franklin diabases (675 Ma, 8°N, 167°E; Fahrig *et al.* 1971) and from the Coronation Sills (647 Ma, 1°S, 163°E; Robertson and Baragar 1972) are in very good agreement.

Independent evidence exists for a Late Hadrynian event that affected the Beltian sediments and probably caused the thermal event transitional between high-grade diagenesis and low-grade metamorphism (see Maxwell and

TABLE 2. Magnetic directions of the intermediate components

Specimen	Site	Before TC Decl.	Before TC Incl.	After TC Decl.	After TC Incl.	Trajectory used
3A1	1	258.8	−10.4	258.7	+5.4	10–30 mT
3A2	1	267.7	−11.8	267.3	+3.4	400°–500 °C
3B1	1	272.8	−17.8	271.8	−2.7	10–30 mT
15B	2	281.2	−6.0	281.1	+4.6	210°–500 °C
33	3	88.3	+22.5	81.8	+10.5	80–100 mT
35A	3	80.5	+10.9	80.1	−1.3	80–100 mT
35B	3	242.1	+34.2	242.5	+47.2	80–100 mT
37A	3	51.5	+33.6	49.6	+9.0	80–100 mT
202B	4	329.0	+32.5	323.1	+30.1	80–100 mT
206B	4	316.2	−5.2	317.3	−9.0	80–100 mT
207A	4	250.0	+32.6	250.0	+22.6	80–100 mT
214A	5	276.0	+33.0	271.8	+25.1	20–100 mT
234A	6	262.3	+35.8	266.4	+28.2	50–100 mT
234B	6	246.2	+4.8	246.5	−0.4	400°–500 °C
264B	8	245.0	+26.3	245.3	+16.4	475°–610 °C
265	8	271.2	+20.8	270.2	+11.4	10–70 mT
361B	10	258.6	−5.2	258.6	+6.2	540°–555 °C
374B	11	286.0	+30.2	283.2	+22.0	300°–475 °C
376B	11	263.7	+43.3	261.9	+33.6	560°–568 °C
381B	12	280.0	+49.2	272.7	+36.3	600°–640 °C
382B	12	293.3	+45.0	284.6	+34.2	295°–400 °C
383B	12	316.3	+53.5	300.2	+46.6	600°–640 °C
385A	12	281.7	+32.2	277.5	+19.8	15–100 mT
385B	12	268.9	+37.7	265.5	+23.8	295°–400 °C
433A	12	108.1	+8.9	107.7	−0.9	550°–570 °C
433B	14	92.4	−1.5	92.6	−10.3	550°–570 °C
434A	14	60.4	−25.7	58.5	−34.2	564°–570 °C
434B	14	244.3	+5.9	243.9	+14.6	570°–576 °C
All specimens: ($N = 28$)		269.2	+16.0	267.7	+17.0	
		$k = 6$, $\alpha_{95} = 11.8°$		$k = 10$, $\alpha_{95} = 9.1°$		
All sites: ($N = 11$)		269.2	+14.7	267.9	+15.5	
		$k = 10.7$, $\alpha_{95} = 14.5°$		$k = 23.2$, $\alpha_{95} = 9.6°$		
Pole position:		5.6°N, 162.0°E		5.0°N, 163.5°E		

Hower 1967). Illite produced during this event has been radiometrically dated by K/Ar and Rb/Sr methods by Goldich *et al.* (1959) as roughly 750 Ma old.

On the Thermal History of the Beltian Sediments in Glacier National Park

It is of interest to speculate about the intensity and nature of the event that produced the Eocambrian partial remagnetization. Eslinger and Savin (1973) have surmised, from $^{18}O/^{16}O$ ratios of coexisting quartz and illite, that isotopic temperatures ranged from 225 °C to 310 °C in the Lower Belt Supergroup during the event. Elimination of the Eocambrian magnetization occurred mostly between 300° and 550 °C (Fig. 4) but sometimes even higher; using theoretical relationships between blocking temperatures and relaxation time calculated by Dunlop (*in* Pullaiah *et al.* 1975) it is estimated that a range of temperatures between 100° and 300 °C could indeed have been responsible for the Eocambrian magnetization, if hematite is the carrier of the magnetization. Hematite appears to be the most likely mineral; ferrous-ferric ratios for the Spokane argillites are very low (Ross 1959, p. 55). The small amount of ferrous iron can be totally accounted for by iron as octahedral cations in illite (Maxwell and Hower 1967, p. 850) or in carbonate minerals, making the presence of more than trace amounts of magnetite quite unlikely. Temperatures of much more than about 200 °C are generally inconsistent with the low metamorphic grade of the argillites, and the lower limit of the inferred isotopic and magnetic temperatures is favored at present.

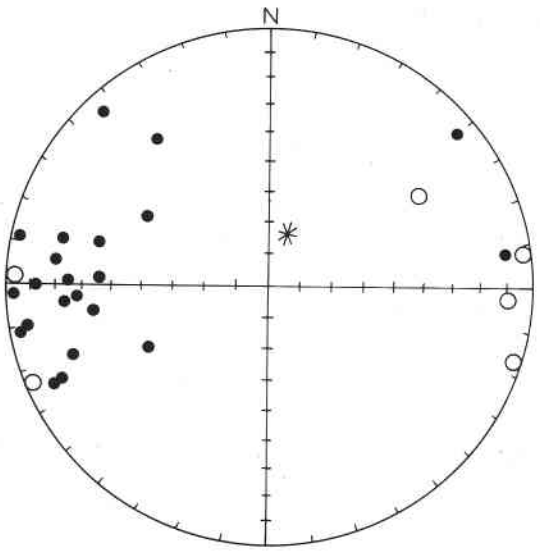


FIG. 5. Equal-area projection of the directions of intermediate coercivity or blocking temperature range. Each dot or circle represents the direction obtained from a specimen by vector subtraction, after tectonic correction. Open symbols pointing upwards, solid symbols pointing downwards. The asterisk is the local present-day field direction.

Eslinger and Savin (1973) attributed the elevated temperatures to burial of the sediments with an overburden of about 7000 m, following Ross (1959). This would imply a temperature gradient of about 30°–45 °C per km, which is not impossible. However, Ross also argued that this overburden was not eroded away until post-Laramide time (Ross 1959, plate 4). If this is so, it is puzzling that the Eocambrian magnetization has persisted, since it is not the burial but rather the removal of the overburden that will leave the magnetic imprint.

Thus we are led to seek a specifically Eocambrian cause for the thermal event. At least two geological processes could have had associated thermal fronts. The first is the East Kootenay orogeny (White 1959) approximately 750 to 850 Ma ago or even earlier (Armstrong 1975), which has been documented in British Columbia. However, effects of this orogeny are not obvious in Glacier National Park and adjacent areas (Ross 1959, p. 75). Subsequent to the orogeny, Late Hadrynian sediments were deposited in a roughly north-south trending zone from Arizona to British Columbia, where they comprise the Windermere Group. In an area adjacent to Glacier National Park these sediments have abundant intercalated volcanics. In the park itself some mafic intrusive rocks (sills) have been dated as 750 Ma old (Obradovich and Peterman 1968). Stewart (1972) has postulated that the volcanics and the Late Hadrynian sedimentary wedge are due to a continental breakup at that time. It is well known that such a process may be accompanied by anomalously high heat flow (*e.g.*, Chapman and Pollack 1975). Thus we conclude that the magnetic minerals of the Beltian sediments in Glacier National Park may have recorded the thermal event associated with this rifting and breakup, though we cannot presently exclude the possibility that the East Kootenay orogeny has had a far-reaching effect into northwestern Montana. The proposed rift was situated to the west of the Glacier Park area, thus leaving the area as part of the North American craton as is suggested by the consistency of the 'Elsonian poles'.

Conclusions

The Spokane Formation has yielded a geomagnetic pole position, based on high-temperature magnetizations, that agrees well with contemporaneous Helikian ('Elsonian') poles from different parts of the North American craton. In addition these sediments recorded a Late Hadrynian magnetization that appears to be associated with low-grade metamorphism changing the claystones to argillites. The Helikian magnetization residing in fine-grained hematite has survived this remagnetization.

A possible cause for the Eocambrian thermal event may have been rifting and continental breakup as advocated by Stewart (1972), though the slightly older East Kootenay orogeny cannot be excluded from consideration.

Acknowledgements

The authors thank the National Park Service for allowing us to collect within its domain. We thank Drs Sam M. Savin, John Stewart, and David T. A. Symons for their comments. Our colleagues, Henry N. Pollack, Bruce R. Clark, Eric J. Essene, John R. Bowman, and Fred J. Mauk, are acknowledged for their helpful discussions and critical reading of the manuscript. We are also indebted to Walter Fahrig and Pierre Lapointe for several constructive criticisms. Financial support was obtained from National Science Foundation Grant GA-38038.

ARMSTRONG, R. L. 1975. Precambrian (1500 m.y. old) rocks of central Idaho — The Salmon River Arch and its role in Cordilleran sedimentation and tectonics. American Journal of Science, **275-A**, pp. 437–467.

BLACK, R. F. 1963. Palaeomagnetism of part of the Purcell System in southwestern Alberta and southeastern British Columbia. Geological Survey of Canada Bulletin 84, pp. 1–31.

CHAPMAN, D. S. and POLLACK, H. N. 1975. Heat flow and incipient rifting in the central African plateau. Nature, **256**, pp. 28–30.

COLLINSON, D. W. and RUNCORN, S. K. 1960. Polar wandering and continental drift: evidence from paleomagnetic observations in the United States. Geological Society of America Bulletin, **71**, pp. 915–958.

ESLINGER, E. V, and SAVIN, S. M. 1973. Oxygen isotope geothermometry of the burial metamorphic rocks of the Precambrian Belt Supergroup, Glacier National Park, Montana. Geological Society of America Bulletin **84**, pp. 2549–2560.

EVANS, M. E., BINGHAM, D. K., and MCMURRY, E. W. 1975. New paleomagnetic results from the Upper Belt–Purcell Supergroup of Alberta. Canadian Journal of Earth Sciences, **12**, pp. 52–61.

FAHRIG, W. F., IRVING, E., and JACKSON, G. D. 1971. Paleomagnetism of the Franklin diabases. Canadian Journal of Earth Sciences, **8**, pp. 455–467.

GOLDICH, S. S., BAADSGAARD, H., EDWARDS, G., and WEAVER, C. E. 1959. Investigations in radioactivity-dating of sediments. American Association of Petroleum Geologists Bulletin, **43**, pp. 654–662.

GRAHAM, J. W. 1949. The stability and significance of magnetism in sedimentary rocks. Journal of Geophysical Research, **54**, pp. 131–167.

HARRISON, J. E. 1972. Precambrian Belt basin of northwestern United States: its geometry, sedimentation, and copper occurrences. Geological Society of America Bulletin, **83**, pp. 1215–1240.

IRVING, E. and MCGLYNN, J. C. 1976. Proterozoic magnetostratigraphy and the tectonic evolution of Laurentia. Philosophical Transactions of the Royal Society of London A, **280**, pp. 433–468.

MAXWELL, D. T. and HOWER, J. 1967. High-grade diagenesis and low-grade metamorphism of illite in the Precambrian Belt series. American Mineralogist, **52**, pp. 843–857.

NORRIS, D. K. and BLACK, R. F. 1961. Application of paleomagnetism to thrust mechanics. Nature, **192** (4806), pp. 933–935.

OBRADOVICH, J. D. and PETERMAN, Z. E. 1968. Geochronology of the Belt Series, Montana. Canadian Journal of Earth Sciences, **5**, pp. 737–747.

PULLAIAH, G., IRVING, E., BUCHAN, K. L. and DUNLOP, D. J. 1975. Magnetization changes caused by burial and uplift. Earth and Planetary Science Letters, **28**, pp. 133–143.

ROBERTSON, W. A. and BARAGAR, W. R. A. 1972. The petrology and paleomagnetism of the Coronation Sills. Canadian Journal of Earth Sciences, **9**, pp. 123–140.

ROSS, C. P. 1959. Geology of Glacier National Park and the Flathead region, northwestern Montana. United States Geological Survey, Professional Paper 296, pp. 1–125.

RUNCORN, S. K. 1964. Paleomagnetic results from Precambrian sedimentary rocks in the western United States. Geological Society of America Bulletin, **75**, pp. 687–704.

SPALL, H. 1971. Precambrian apparent polar wandering: evidence from North America. Earth and Planetary Science Letters, **10**, pp. 273–280.

STEWART, J. H. 1972. Initial deposits in the Cordilleran geosyncline: evidence of a Late Precambrian (< 850 m.y.) continental separation. Geological Society of America Bulletin, **83**, pp. 1345–1360.

WHITE, W. H. 1959. Cordilleran tectonics in British Columbia. American Association of Petroleum Geologists Bulletin, **43**, pp. 60–100.

ZIJDERVELD, J. D. A. 1967. AC demagnetization of rocks: analysis of results. *In*: Methods in paleomagnetism (D. W. Collinson, K. M. Creer, and S. K. Runcorn (*Eds*.)). Elsevier, New York, pp. 254–286.